

TaxaExpect

Estimate spatially-explicit Bayesian priors for expected species composition. Part of the TaxaID ecosystem.

A taxonomic assignment based on a match to a reference can propose an implausible species. These are sometimes accepted as fact, but often are screened afterwards, or prevented from occurring by reducing the reference list to known local species.

Bayes' Theorem improves taxonomic assignment by considering, before any match score is seen, how large a share of the detections at the sampling location a hypothesized taxon is expected to account for. TaxaExpect estimates these priors from occurrence records. This is typically GBIF (Global Biodiversity Information Facility; GBIF Secretariat, Copenhagen, Denmark) records fetched via TaxaFetch, or user-supplied data. Although occurrence data are often sparse and biased, they are usually sufficient to distinguish among taxa with similar match scores but very different geographic ranges. TaxaExpect's priors are compositional shares – the expected relative share of a species at a site, not an occurrence probability – estimated directly from occurrence data rather than from an occurrence/occupancy model.

TaxaExpect uses a **site-centered distance kernel**: each species' prior is its kernel-weighted share of related, nearby occurrence records, computed directly at the query site. Weight decays smoothly with distance (and, optionally, with a covariate such as depth), so spatial borrowing degrades continuously rather than switching on or off at an arbitrary cell boundary, and even lightly-sampled sites get informative priors by shrinking toward the surrounding region's composition. The kernel bandwidth is chosen from data via leave-one-block-out composition prediction (`calibrate_kernel_bandwidth()`), not guessed, and the resulting prior field can be mapped continuously over the study area with `plot_theta_surface()`. For taxa never reported in the study area, TaxaExpect prices a “dark diversity” prior from a Good-Turing/Chao- anchored presence-distance curve.

The resulting spatial priors are helpful in reducing false positives from ecologically implausible assignments and can break ties between similar-scoring species, rescuing species-level resolution that would otherwise be lost to defensive upranking.

Overview

TaxaExpect generates theta priors for taxonomic assignment from occurrence data. Theta is compositional: the expected relative share of a species at a site, $P(\text{a random legitimate detection} = \text{species X})$. It is not an occurrence probability, and not occupancy – see *Shared detection effort* below for the formal definition. The current (kernel) pathway estimates expected species composition directly at a site via distance-weighted occurrence sharing, incorporating habitat stratification and, optionally, a covariate such as depth.

Priors are organized into three branches (`prior_branch`):

- **resident_observed** – species with kernel-weighted local occurrence evidence (direct estimate)
- **resident_undetected** – species plausibly present but not locally recorded (singleton mirrors, a Good-Turing floor, and named claimants priced on a shared presence-distance curve)
- **transport** – domestic/food/cultivar species and other non-resident presence hypotheses

Assumptions

Shared detection effort

TaxaExpect models *relative* abundance: the probability that a randomly selected observation at a site belongs to a given taxon. The denominator `n_total_at_site` is the total count of all observations at a site and serves as the shared effort measure for every taxon in the model.

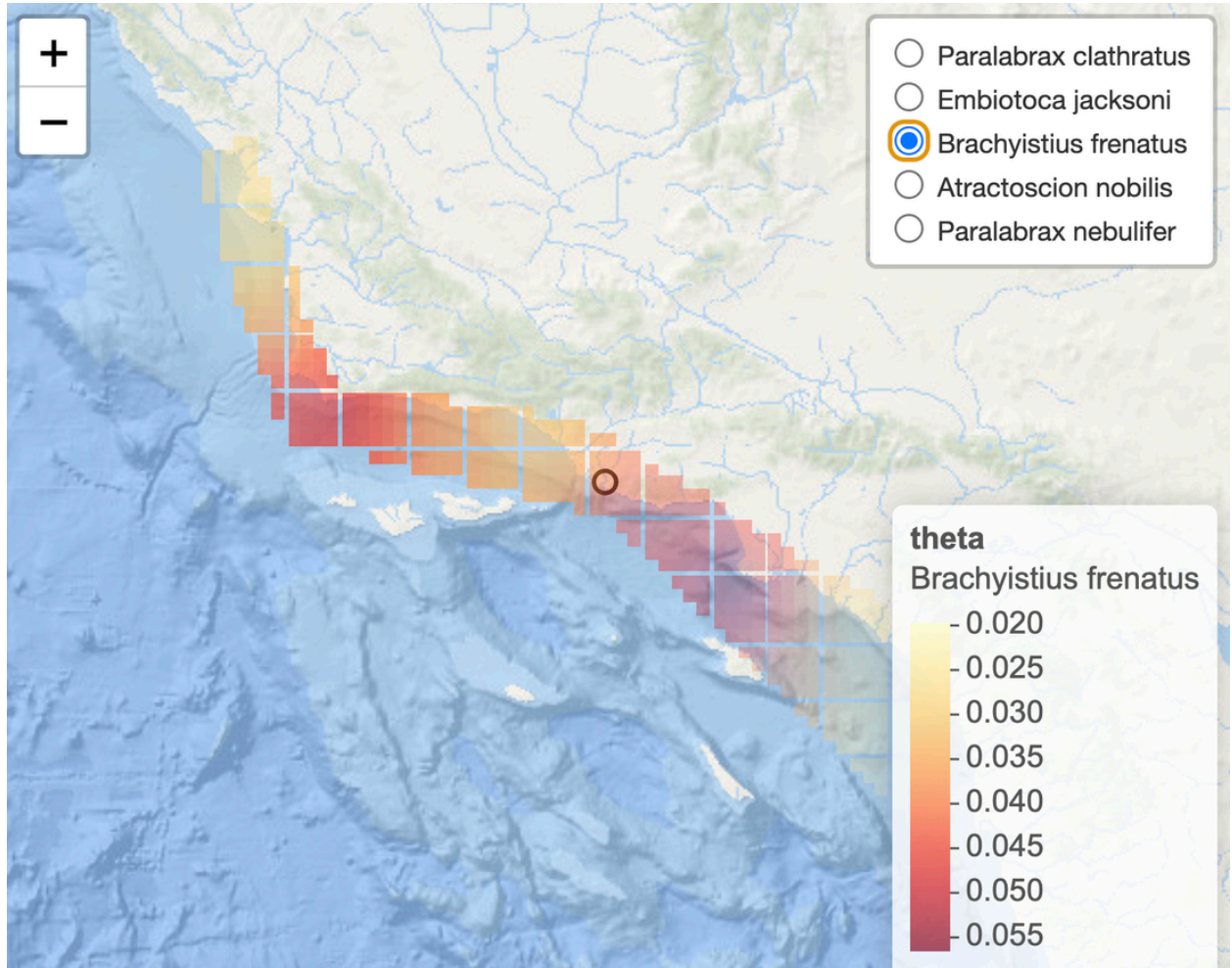


Figure 1: Interactive prior-field map produced by `plot_theta_surface()`. The surface is the estimator evaluated continuously across a lattice, not one value per grid cell, so it shows how a species' expected share changes across the region rather than only at the sampling site. Here θ for kelp surfperch (*Brachyistius frenatus*) rises along the Santa Barbara Channel and the Los Angeles coastline, and the hollow circle marks the focal site. Passing several taxa to `taxon` adds the layer toggle at top right, so priors can be compared species by species. The surface is clipped to a supplied search polygon via `mask`, which is why it stops at the coast instead of painting inland.

All taxa in a particular model should be detected through a similar sampling process. Combining taxa collected by incommensurable methods corrupts the shared denominator: a phytoplankton cell count and a bird point-count sighting are not equivalent detection events. Mixing them implies that plankton-sampling effort informs expected bird relative abundance, which is not true. The two surveys represent independent detection processes.

Problematic mixing (do not combine in a single model):

- Phytoplankton cell counts + bird point counts
- eDNA reads from different gene markers (e.g. 12S fish reads combined with COI invertebrate reads; amplification efficiency differs between markers, so read counts are not on a shared effort scale)
- Arthropods from pitfall traps + arthropods from Malaise traps

Valid pooling:

- All fish species from the same eDNA marker on the same filter
- All bird species detected during the same standardized point count
- All macroinvertebrate taxa from the same kick-net sample

This grouping is not automatic, and there is no automated way to make it automatic. `estimate_kernel_priors()` has no way to tell, on its own, which taxa share a detection process – if you leave `sampling_group_col` unset, it silently pools every taxon into one shared denominator regardless of detection process, with no warning and no error. Supplying a correct grouping is your responsibility, and it must be done by hand, from real knowledge of how each taxon was detected – there is no way to infer “comparable detection method” from taxonomy or occurrence data alone (two co-occurring taxa sampled by different gear look identical in an occurrence table). Automatic, data-driven classification of detection process does not work reliably for this: tested against a real, full-scale expert-classified dataset, it answered a different question than `sampling_group` is meant to answer (whether a group clears a per-site record-count floor, not whether it shares a detection process) – it fragmented a single real detection process into dozens of spurious groups, and separately conflated two taxa an expert had deliberately kept apart (a genuine marine survey target and likely airborne contamination) purely because neither cleared the floor on its own. An LLM guess at the grouping is not a substitute for real methodological knowledge either, for the same reason: the distinction lives in how the data were collected, not in anything visible in the records themselves. Two options remain: 1) manually choose similar taxa that are sampled in similar ways, the approach this project’s own analysts have used successfully for every real dataset handled so far; or 2) build separate models for each survey type and pass the appropriate priors to TaxaAssign for the relevant taxonomic hypotheses.

Choosing a grouping by hand does not require worrying about sample size. `estimate_kernel_priors()` does **not** need pre-merged, sample-size-adequate groups for statistical adequacy – it produces honestly wide uncertainty for a small group on its own (confirmed on a real 9-group expert classification, including a single-taxon group and a 3-taxon group, both of which fit cleanly with appropriately wide `theta_sd`). Classify at whatever granularity genuinely reflects distinct detection methods, and let the estimator’s own uncertainty reflect how much data backs each group – there is no separate merging step to get right.

Installation

```
# Requires TaxaTools (foundation package)
devtools::install("path/to/TaxaTools")
devtools::install("path/to/TaxaExpect")
```

Quick Start

```
library(TaxaExpect)

# 1. Calibrate the kernel bandwidth (and regional back-off m) by
#    leave-one-block-out composition prediction -- never hand-set.
calib <- calibrate_kernel_bandwidth(
  occurrence_data = occurrences, # habitat-labelled occurrence records
  site_habitat    = "Marine",
  lambda_grid     = c(10, 25, 50, 100, 200) # km
)
lambda_km <- calib$best$lambda_km

# 2. Estimate kernel priors at your site (resident_observed rows)
kernel_fit <- estimate_kernel_priors(
  occurrence_data = occurrences,
  site_lat        = 34.45,
  site_lon        = -120.47,
  site_habitat    = "Marine",
  lambda_km       = lambda_km
)

# 3. Add resident_undetected rows (singleton mirrors + Good-Turing floor)
undetected <- generate_undetected_diversity(kernel_fit)

# 4. Assemble the prior table for TaxaAssign
priors <- dplyr::bind_rows(kernel_fit$priors, undetected)

# 5. Explore the prior field
plot_theta_surface(kernel_fit, occurrence_data = occurrences,
  taxon = "Girella nigricans")
```

Key Functions

Kernel pathway (current, recommended)

Calibration: - `calibrate_kernel_bandwidth()` – choose the geographic bandwidth (and optional covariate bandwidth, and the regional back-off mass m) by leave-one-block-out composition prediction

Prior estimation: - `estimate_kernel_priors()` – site-centered kernel estimation of `resident_observed` priors (no grid, no model fit) - `generate_undetected_diversity()` – singleton-mirror and global-floor `resident_undetected` priors (accepts kernel or GLMM input) - `generate_presence_curve_evidence()` / `generate_user_specified_evidence()` + `apply_undetected_evidence(pricing = "curve")` – price named unobserved claimants (regional, watch-listed, or distance-clamped) on a shared presence-distance curve - `condition_evidence_on_habitat()` – multiply any evidence table’s presence weights by each taxon’s weight for the site habitat (from the cached LLM habitat lookup), floored at the zero-evidence clamp, so evidence obeys the same habitat stratification as the resident priors while habitat bleed survives in proportion - `generate_domestic_food_priors()` – transport-branch priors for domestic, food, and cultivar species

Diagnostics and reporting: - `plot_theta_surface()` – continuous prior-field map, evaluating the estimator on a lattice via FFT - `kernel_budget_sensitivity()` – reports how the Good-Turing budget behind `theta_present` moves across counting radius/bandwidth choices - `report_priors()` – generate report section for `assemble_report()`

Statistical Methods

TaxaExpect's current pathway prices each species' prior as its kernel-weighted share of occurrence records in the focal-habitat stratum, shrunk toward the regional composition by m pseudo-records (a Dirichlet back-off):

$$\text{theta}_i = (\text{c}_i * s + m * \text{p}_i) / (\text{n_eff} + m)$$

where c_i is species i 's summed distance-weighted record count ($\text{w}_r = \exp(-\text{d}_r/\text{lambda_km})$, optionally multiplied by a covariate factor such as depth), $s = \text{n_eff} / W$ is an effective-scale factor, p_i is the unweighted regional (habitat-stratified) record share, and $\text{n_eff} = W^2 / \sum(\text{w}_r^2)$ is the Kish (1965) effective sample size of the weighted neighborhood. $\text{alpha} = \text{c}_i * s + m * \text{p}_i$ and $\text{beta} = (\text{n_eff} + m) - \text{alpha}$ are the row's Beta parameters directly – no delta-method back-transformation, no phi cap/floor, no Jeffreys fallback are needed, since alpha/beta are built from non-negative counts and pseudo-counts by construction and cannot produce the boundary pathologies a link-scale model can. lambda_km (and, if used, a covariate bandwidth and m) is chosen from data by `calibrate_kernel_bandwidth()`'s leave-one-block-out composition prediction – never hand-set.

For undetected species, a Good-Turing/Chao-anchored presence-distance curve (`apply_undetected_evidence(pricing = "curve")`) prices each named claimant as a two-point presence mixture, $\text{theta} = \text{w} * \text{theta_present}$, where theta_present is the neighborhood's own observed singleton mean and w comes from the claimant's distance to its nearest occurrence record (or, for iNaturalist range evidence, from independent verification).

Real validation. Leave-one-block-out testing – holding out blocks of occurrence records and scoring composition predictions against them by multinomial log-loss – found that single-cell GLMM prediction scored *worse* than ignoring space entirely, while the kernel estimator beat both regional pooling and the single-cell predictor at every bandwidth tested. On real Great Lakes data, switching from the GLMM baseline to the kernel estimator raised species co-detections from 237 to 564 and precision from 0.748 to 0.868 (independently validated against a held-out checklist).

For the full statistical derivation, assumptions, and references, see `inst/TaxaExpect_supplemental_methods.md`.

Part of TaxaID

TaxaExpect receives habitat-annotated occurrence data from TaxaHabitat and produces spatially-explicit priors for TaxaAssign (posterior computation).

Ecosystem: TaxaFetch -> TaxaHabitat -> **TaxaExpect** -> TaxaAssign

See the TaxaID README for ecosystem overview and installation instructions.

Citation

Lafferty, K.D., 2026, TaxaID – A modular R ecosystem for Bayesian taxonomic assignment: U.S. Geological Survey software release, <https://doi.org/10.5066/xxxxxx>.

Software Requirements

- R ($\geq 4.1.0$; R Core Team 2025)
- TaxaTools (foundation package)
- TaxaFetch, TaxaHabitat (for fetching and habitat-labeling occurrence records upstream of the kernel pathway; in Suggests)

All dependencies are declared in the DESCRIPTION file and installed automatically.

Developed with Claude Code (Anthropic PBC, San Francisco, California).

References

- Chao, A. (1984). Nonparametric estimation of the number of classes in a population. *Scandinavian Journal of Statistics*, 11(4), 265–270.
- Good, I.J. (1953). The population frequencies of species and the estimation of population parameters. *Biometrika*, 40(3-4), 237–264.
- Kish, L. (1965). *Survey Sampling*. New York: Wiley.
- R Core Team (2025). R: A Language and Environment for Statistical Computing. V.4.5.2. R Foundation for Statistical Computing, Vienna, Austria. <https://www.r-project.org>